EVT 3D-PRINTED HOUSING - GENERAL ARRANGEMENT

Controlled build-to-print option for the EVT lot. Upper electronics and power modules are separate sealed volumes. Four PCB-MIC pods are replaceable. The power base accepts the RELiON RB40 envelope; the power lid provides PCB-PWR and Victron MPPT mounting provisions.

UPPER ELECTRONICS BASE

150 x 115 x 44

UPPER LID

150 x 115 x 12 total

POWER BASE

215 x 184 x 196

POWER LID

215 x 184 x 13 total

Controlled interfaces

- PCB-MAIN: 110 x 75 x 12 PCBA envelope, M3 pattern X5/105 Y5/70 mapped to four internal bosses.
- PCB-PWR: 90 x 60 mm, four M3 mounting points, >=20 mm connector service volume retained.
- Battery: RB40 197 x 166 x 173 mm inside 205 x 174 x 190 mm cavity; terminal volume continues into lid.
- RF: three SMA bulkhead holes, 42 mm pitch; CAB.0243 minimum bend radius 6.8 mm.
- Acoustic: centered 2.0 mm free opening, adhesive exclusion radius 1.5 mm, membrane in removable cap.

MIC POD BASE x4

30 x 28 x 14

MIC POD CAP x4

30 x 28 x 3

EVT 3D-PRINTED HOUSING - MANUFACTURING AND INSPECTION

Print controls

- Printer class: FlashForge Adventurer 5M Pro, build volume 220 x 220 x 220 mm.
- Material: UV-stabilized ASA; 0.4 mm nozzle; 0.20 mm layers; 5 perimeters; 35 percent gyroid infill.
- Power base maximum print envelope 215 x 184 x 196 mm. No brim; verify actual calibrated build area before start.
- Do not place supports or post-process tools on gasket faces, acoustic openings, insert bores or RF seats.
- Condition parts 24 h before dimensional inspection and insert installation.

First article inspection - release remaining lot only after PASS

DIMENSION	All outer envelopes +/-0.5; PCB and OTS item fit without force; service clearances retained.
FASTENERS	Insert pull and torque sample; M3 <=0.4 N m at PCB; M4 torque set by hardware stack.
SEAL	Uniform 15-25 percent gasket compression; visual light-gap check followed by controlled IP test.
ACOUSTIC	Four open ports; no adhesive/paste/coating; gain, phase, noise and delay within EVT limits.
RF	Bulkhead seating and torque, cable bend/retention, installed VSWR/sensitivity/coexistence.
POWER	Battery terminal cover and tool clearances, gland pull, fuse access, temperature and fault checks.
SERVICE	USB-C, SIM, microSD and SWD access without PCB or seal damage; cover can be resealed.
TRACEABILITY	Part/revision/material lot/printer/profile/operator/date recorded for every printed set.

Release boundary

These files authorize fabrication of the EVT printed housing set. Physical IP, UV/environmental, structural, thermal, RF and acoustic results are collected on the first articles and block serial transfer. Do not scale STL meshes or change wall, hole, gasket or acoustic geometry without an ECO.